Printed Pages:02 **Sub Code: RME -503**

Paper Id: 232048 **Roll No.** | | | | | | | | | | |

B. TECH
(SEM-V) THEORY EXAMINATION 2022-23
MANUFACTURING SCIENCE & TECHNOLOGY

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2x7 = 14

(a) Discuss about single point cutting tool used in machining.
(b) Discuss about coolant used in Lathe machine.
(c) Define straight polarity?
(d) Explain about oxidizing flame used in oxy acetylene gas welding.
(e) Explain: a) boring b) Reaming
(f) Define air aspiration welding defect.
(g) Discuss the types of wear mechanism in tools.

SECTION B

2. Attempt any *three* of the following: 7x3 = 21

(a) Discuss the tool geometry of a single point cutting tool and draw neat diagram.
(b) Discuss about milling machine. Also recommend the application of milling machine in industry.
(c) Discuss about Centreless grinding machine. Also mention their advantages and limitations.
(d) Illustrate the spot resistance welding. And explain the main factors to be considered in resistance welding?
(e) With the help of a neat sketch, explain the working principle and machine setup of Abrasive Jet Machining (AJM). Also discuss the effect of process variables on material removal rate in AJM.

SECTION C

3. Attempt any *one* part of the following: 7x1 = 7

(a) Discuss the various types of chips? And explain under what condition each is formed.
(b) A carbide tool with MS work piece was found to give tool life of 2 hrs while cutting at 0.4 m/min. compute the tool life if the same tool is used at speed of 24% higher than previous one. Also determine the value of cutting speed if the tool is required to have tool life of 3 hrs. Assume taylors exponent n to be 0.2?

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4. Attempt any *one* part of the following: 7x1 = 7

(a) Compare between up milling and down milling. Also write their use in industry.

(b) Describe the mechanism of a dividing head and explain how it is used for indexing a work-piece e.g., a gear blank.

5. Attempt any *one* part of the following: 7x1 = 7

(a) Discuss the through feed and Infeed in grinding.

(b) Illustrate grinding wheel specification? Clearly differentiate between grade and structure of a Grinding wheel.

6. Attempt any *one* part of the following: 7x1 = 7

(a) Explain the classification of welding? Also illustrate different types of welding defects.

(b) Illustrate the Metal Inert gas (MIG) welding with neat sketch.

7. Attempt any *one* part of the following: 7x1 = 7

(a) Discuss the principal of Plasma arc machining (PAM). Giving neat sketch, and explain the working of plasma arc machining process.

(b) Illustrate the working of Electro chemical Machining (ECM) process. What is the function of electrolyte in ECM? Enlist the application of ECM.

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